

THE OPERATING SYSTEM FOR LEVEL-4 AUTONOMOUS MOBILITY IN GNSS-DENIED ENVIRONMENTS

Modular, deterministic Level-4 autonomous retrofit kits for electric passenger fleets and tactical defense platforms at 70% lower CAPEX.

Trustia AI Technologies

Dual-Use Autonomous Mobility & Defense Robotics Platform

LEAD ARCHITECT Murat Furkan Bayram (17, 80%) 16k Lines Custom Autonomy Stack	EU REGISTRATION PIC: 861711529 • EIT: CUS15554 Official European Commission Partner	VERIFICATION 1,301 / 1,301 Tests Passing (100%) Zero Third-Party Dependencies
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THE PROBLEM: WHY COMMERCIAL ROBOTAXIS ARE STALLED

Legacy Bespoke Approaches Suffer from Triple Scalability Roadblocks

TRUSTIA AI • Global Pitch Deck

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1. \$250k+ Build CAPEX

Waymo and Cruise manufacture custom vehicles from scratch costing \$250,000 to \$350,000 each. Fleet operators cannot scale economically with 7+ year payback horizons.

2. GPS Fragility & Spoofing

Modern electronic warfare, urban skyscraper canyons, and tunnels disable standard satellite navigation. Unmanned systems drift and stall without satellite fix.

3. Hardware OEMs Lack Brains

Vehicle manufacturers engineer outstanding mechanical chassis, but lack the proprietary SLAM, kinematic path planners, and functional safety layers required for Level-4.

THE SOLUTION: MODULAR RETROFIT KITS AT 70% LOWER COST

Converting Mass-Production EVs (Hyundai Ioniq 5) in 48 Hours with Zero Body Damage

Technology Pillar	Architectural Implementation	Operational Metric
GNSS-Denied 3D SLAM	Ouster 128-LiDAR + IMU pose-graph optimization	5 cm accuracy in tunnels & skyscraper canyons (\$0 GPS cost)
Kinematic Hybrid A*	Ackermann dynamics solved under 15ms loop	Guaranteed collision-free trajectories respecting slip angles
Drive-by-Wire Bridge	Kvaser CAN-FD directly injecting LKAS & SCC commands	5ms driver override reaction, 100Hz control loop
ASIL-D Functional Safety	ISO 26262 Minimal Risk Maneuver (MRM) fallback	Automatic safe-stop in under 5 seconds upon sensor fault

TECHNICAL MOAT: 1,301 PASSING TESTS & ZERO-DEPENDENCY CORE

Audited Deterministic C++17/20 Architecture Free from Foreign Black Box Risks

16,000+

Lines of Proprietary C++/Python

1,301

Automated Unit & Integration Tests (100%)

< 15 ms

Deterministic Real-Time Cycle Time

- **Zero Third-Party Dependency:** Handcrafted algorithmic pipeline eliminates supply-chain vulnerabilities, open-source CVE exploits, and foreign backdoors.
- **NATO STANAG 4586 Level 4 & SAE AS6091:** Tactical dual-use capability allows instant deployment onto military UGVs and border defense platforms.
- **Tactical C2 Command Console:** Live desktop GUI with MIL-STD-2525 symbology for fleet tele-operation and remote intervention.

HARDWARE ECOSYSTEM: HYUNDAI IONIQ 5 27-PART LEVEL-4 BOM

World-Class Sensor Configuration Verified by Top Distributors (\$23.8k Total BOM)

Subsystem	Model & Technical Spec	Unit Cost
Roof 3D LiDAR	Ouster OS2-128 Rev 7 (128 Channels, 200m+ range, 360°)	\$7,270 (₺351k)
Blind-Spot LiDARs (2x)	Livox Mid-360 Solid-State (Front bumper corners)	\$4,820 (₺232k)
Central AI Compute	NVIDIA Jetson AGX Orin 64GB (275 TOPS) + Saeed J501 Carrier	\$3,760 (₺181k)
77 GHz Radars (2x)	Continental ARS 408-21 (All-weather fog & dust penetration)	\$1,230 (₺59k)
RTK GNSS + IMU	Septentrio mosaic-go Dual-Antenna Heading + Industrial IMU	\$1,080 (₺52k)
Automotive Cameras (4x)	Leopard Sony IMX390 HDR GMSL2 Cameras (Traffic lights & signs)	\$2,040 (₺98k)
CAN-FD + 5G Router	Kvaser U100 CAN-FD + Teltonika RUTX50 Dual-SIM 5G/V2X	\$1,230 (₺60k)
Mounting, PDU & Harness	Flush roof rack, ELO relays, Mean Well DC-DC regulator	\$2,370 (₺115k)
TOTAL HARDWARE BOM	27-Part Complete Turnkey Level-4 Conversion Kit	\$23,800 (₺1.15M)

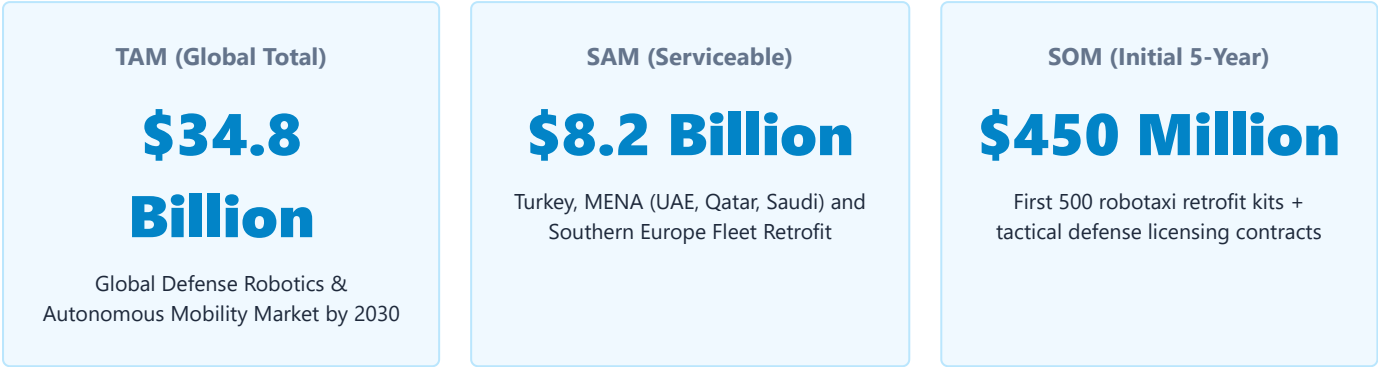
BUSINESS MODEL: DUAL-LAYER HARDWARE KIT & HIGH-MARGIN AAAS

Upfront Kit Gross Margin Plus Recurring Software Subscription

Tier 1: Civil L4 Kit \$18,500 BOM: \$11,500 • \$250/mo AaaS Campus shuttles, airport transfers, enclosed industrial logistics.	Tier 2: Robotaxi Turnkey \$35,000 BOM: \$22,800 • \$450/mo or \$0.18/km 14-Month Payback (ROI)! Commercial taxi fleets, ride-hailing networks.	Tier 3: NATO Hardened UGV \$55,000 BOM: \$32,800 • \$1,200/mo AaaS \$250k - \$1M custom PoC contracts. Tactical armed/reconnaissance UGVs.
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MARKET OPPORTUNITY: \$34.8B COMBINED TAM BY 2030

Riding the Structural Shift to Autonomous Commercial Fleets (14.2% CAGR)



COMPETITIVE ADVANTAGE: WHY TRUSTIA WINS

Bypassing Multi-Million Dollar Barriers with Modular Agility

Criterion	Legacy Defense Giants	Open Source (Autoware / Nav2)	TRUSTIA AI
GPS-Denied Precision	\$500k+ Military INS	Unstable, frequent drift	Centimeter 3D SLAM (\$0 GPS)
Retrofit Agility	Multi-year redesign	Fragile package dependencies	48-Hour Plug & Play CAN-FD
Unit Economics	Cost-plus millions	Zero commercial support	\$35k Kit + High Margin AaaS
Verification	Proprietary black box	Untested in real hardware	1,301 Automated Tests (100%)
EU & State Registry	Foreign export restrictions	Uncertified	EU PIC #861711529 • EIT Partner

TRACTION & ACCREDITATION: 85+ GLOBAL APPLICATIONS
(SEPT 2026)

Rapidly De-risked Across European Union, Silicon Valley, MENA & Defense Pipelines

Institution / Entity	Accreditation & Registry Code	Official Status
European Commission	Participant Identification Code (PIC): 861711529	OFFICIALLY REGISTERED
EIT Urban Mobility	Official Partner ID: CUS15554 • €100k Grant (ID: 3.1.02-1206-3732.3)	PARTNER / SUBMITTED
BAYKAR Teknoloji	Level-4 Autonomy & 3D LiDAR SLAM Supplier Application	SUBMITTED & CONFIRMED
DEIK (Foreign Trade)	Digital Technologies Business Council Official Registration	SUBMITTED & CONFIRMED
ASELSAN	Supplier Portal Land Systems Autonomy (No: 0050569CCE941FD1A49FCEFB9B7BE7D6)	PRE-EVALUATION ACTIVE
QSTP (Qatar Foundation)	\$30M Tech Venture Fund + 4-Week Doha Sprint Incubation Program	SUBMITTED (9-PAGE / 8-PAGE)
Z Fellows (Silicon Valley)	Live Zoom Partner Interview with Grace Kasten (Pace Capital)	INTERVIEW: SEPT 17
fonbulucu (SPK Turkey)	15M TL Base / 18M TL Cap / 150M TL Valuation Equity Crowdfunding	PRE-REVIEW (W1MV5K)
Crunchbase Verified	Heat Score 85 ↑, CB Rank 534k, Active \$500k Pre-Seed Round Profile	VERIFIED PROFILE

THE ASK & USE OF FUNDS: \$500,000 PRE-SEED ROUND

SAFE Note (\$5.0M Post-Money Cap, 20% Discount) for 18-Month Operational Runway

45% (\$225k) Hardware & Sensor Fleet 2 Ioniq 5 Test Vehicles	35% (\$175k) Core Engineering Team Embedded, SLAM & ROS2	15% (\$75k) Proving Ground Trials Bilisim Vadisi & Doha	5% (\$25k) IP & ASIL-D Audits PCT Patents & Safety
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Partner with Trustia in Shaping the Sovereign Autonomous Future

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